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5120

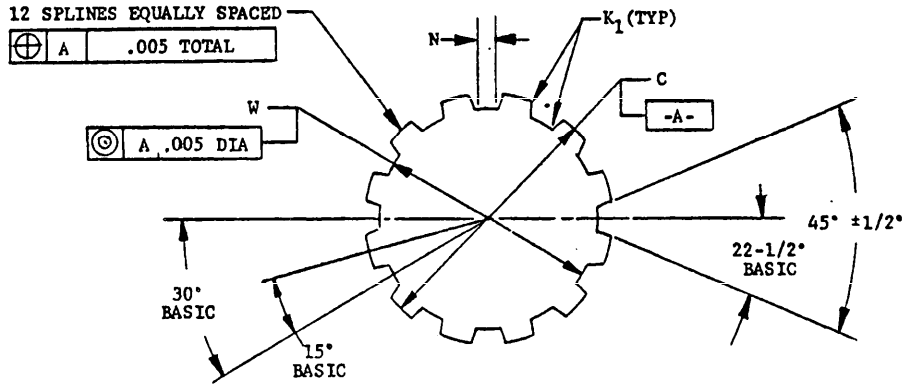


TABLE I - ELEMENT DIMENSIONS

ELEMENT NUMBER	WRENCH SIZE REF	C DIA		W DIA		N		K_1 RADIUS	
		MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX
6	.188	.228	.235	.189	.194	.020	.026	.003	.008
7	.219	.264	.271	.222	.227	.023	.029		
8	.250	.301	.308	.253	.259	.026	.032		
9	.281	.337	.345	.285	.291	.029	.035	.005	.010
10	.312	.373	.381	.318	.324	.031	.037		
12	.375	.447	.456	.383	.389	.037	.043		
14	.438	.522	.531	.448	.455	.043	.050		
16	.500	.595	.605	.514	.521	.048	.055	.005	.015
18	.562	.668	.679	.578	.585	.054	.061		
20	.625	.742	.753	.644	.651	.060	.068		
22	.688	.817	.828	.711	.717	.065	.073	.005	.020
24	.750	.890	.901	.774	.782	.071	.079		
26	.812	.963	.974	.838	.846	.077	.086	.005	.025
28	.875	1.037	1.049	.904	.912	.082	.091		
30	.938	1.112	1.124	.970	.978	.088	.097		
32	1.000	1.185	1.197	1.034	1.042	.094	.103	.010	.030
34	1.062	1.258	1.270	1.099	1.107	.099	.108		
36	1.125	1.333	1.345	1.165	1.173	.105	.114		
38	1.188	1.406	1.419	1.229	1.239	.111	.120	.010	.035
40	1.250	1.480	1.493	1.293	1.303	.116	.125		
42	1.312	1.553	1.566	1.358	1.368	.122	.131		
44	1.375	1.628	1.641	1.424	1.434	.127	.136	.010	.040
46	1.438	1.702	1.715	1.489	1.499	.133	.142		
48	1.500	1.775	1.788	1.554	1.564	.139	.148		
50	1.562	1.849	1.862	1.619	1.629	.144	.154	.015	.045
52	1.625	1.924	1.937	1.685	1.695	.150	.160		
54	1.688	1.998	2.011	1.750	1.760	.156	.166		
56	1.750	2.072	2.085	1.815	1.825	.161	.171		
58	1.812	2.145	2.158	1.880	1.890	.167	.177	.015	.050
60	1.875	2.219	2.233	1.945	1.955	.173	.183		
62	1.938	2.293	2.307	2.011	2.021	.178	.189		
64	2.000	2.367	2.381	2.076	2.086	.184	.195		
66	2.062	2.440	2.454	2.141	2.151	.189	.200		
68	2.125	2.514	2.528	2.206	2.216	.195	.206		
70	2.188	2.589	2.603	2.271	2.281	.201	.212		

NOTES:
1. TOLERANCES SHALL NOT BE CUMULATIVE.
2. DIMENSIONS IN INCHES UNLESS OTHERWISE SPECIFIED.
3. THE GOVERNMENT HAS A ROYALTY FREE LICENSE UNDER U.S. PATENT 3,354,757 FOR THE BENEFIT OF MANUFACTURERS OF THIS WRENCHING ELEMENT EITHER FOR THE GOVERNMENT OR FOR USE IN EQUIPMENT TO BE DELIVERED TO THE GOVERNMENT.

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(D) DENOTES CHANGES

APPROVED 6 MAY 1969 REVISED (A) 3 DEC 73 (B) 15 AUG 1977 (C) 6 DECEMBER 1978 (D) 18 APR 1984

USER SYMBOLS:

REVIEWER SYMBOLS:
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P.A.
NAVY - AS
Other Cust

USAF - 11
ARMY - AV

TITLE

WRENCHING ELEMENT, EXTERNAL SPLINE, DIMENSIONS FOR

MILITARY STANDARD

MS33787

PROCUREMENT SPECIFICATION

SUPERSEDES:

SHEET 1 OF 2

DD FORM 1 MAR 72 672-1 (Coordinated)

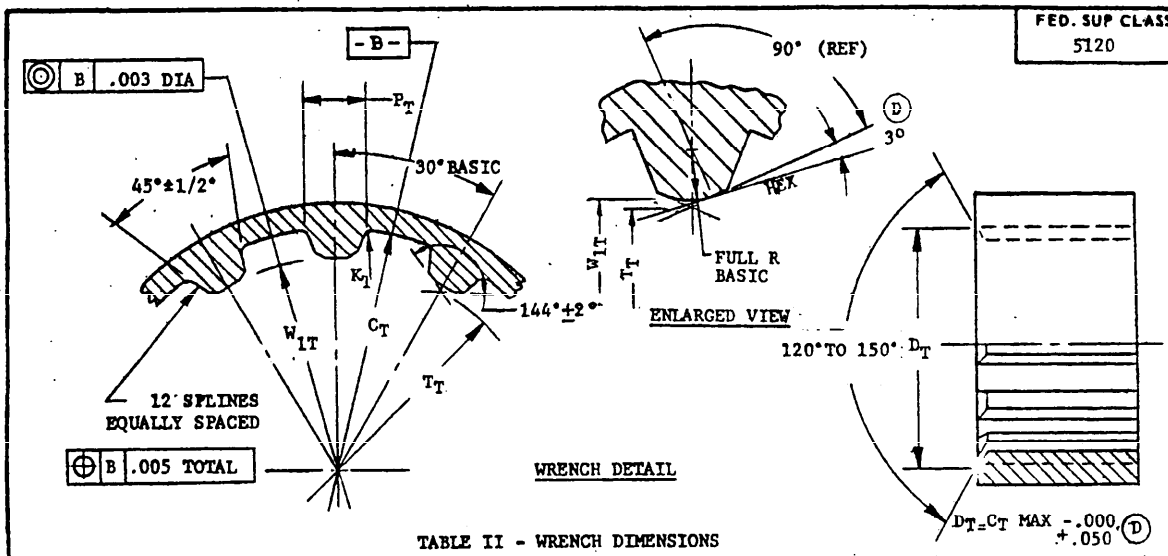
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WRENCH DETAIL

TABLE II - WRENCH DIMENSIONS

WRENCH NO.	WRENCH SIZE REF	W_{1T} DIA		C_T DIA		(D) P_T		T_T		K_T RADIUS		FULL R
		MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	BASIC
6	.188	.198	.202	.240	.244	.029	.036	.1967	.2008			.013
7	.219	.229	.234	.276	.280	.033	.040	.2278	.2319			.015
8	.250	.263	.267	.313	.317	.038	.045	.2609	.2650			.018
9	.281	.295	.299	.350	.354	.043	.050	.2930	.2971	.005	.010	.020
10	.312	.329	.334	.386	.390	.046	.054	.3271	.3313			.022
12	.375	.394	.398	.461	.465	.055	.063	.3913	.3955			.026
14	.438	.458	.463	.536	.541	.064	.073	.4555	.4597			.031
16	.500	.525	.531	.610	.615	.073	.082	.5218	.5270			.035
18	.562	.590	.596	.684	.689	.082	.092	.5860	.5911			.039
20	.625	.655	.662	.758	.764	.091	.101	.6512	.6574			.044
22	.688	.721	.728	.833	.839	.099	.110	.7164	.7226	.010	.015	.048
24	.750	.786	.795	.906	.912	.108	.119	.7816	.7889			.053
26	.812	.852	.862	.979	.986	.116	.128	.8469	.8551			.057
28	.875	.917	.926	1.054	1.061	.126	.138	.9110	.9193			.061
30	.938	.983	.994	1.129	1.136	.135	.147	.9773	.9866			.066
32	1.000	1.048	1.059	1.202	1.209	.144	.156	1.0415	1.0508			.070
34	1.062	1.112	1.125	1.275	1.282	.153	.165	1.1057	1.1160			.074
36	1.125	1.179	1.191	1.350	1.357	.163	.175	1.1719	1.1823			.079
38	1.188	1.245	1.255	1.424	1.431	.172	.184	1.2372	1.2454			.083
40	1.250	1.309	1.323	1.498	1.505	.181	.193	1.3013	1.3127	.010	.020	.088
42	1.312	1.375	1.387	1.571	1.578	.191	.203	1.3666	1.3759			.092
44	1.375	1.441	1.454	1.646	1.653	.200	.212	1.4318	1.4432			.096
46	1.438	1.506	1.520	1.720	1.727	.209	.221	1.4970	1.5084			.101
48	1.500	1.571	1.586	1.793	1.800	.218	.230	1.5612	1.5736			.105
50	1.562	1.635	1.650	1.867	1.875	.227	.240	1.6254	1.6378			.109
52	1.625	1.702	1.717	1.942	1.950	.236	.249	1.6916	1.7041			.114
54	1.688	1.767	1.783	2.016	2.024	.245	.258	1.7558	1.7693			.118
56	1.750	1.833	1.850	2.090	2.098	.255	.268	1.8221	1.8355			.123
58	1.812	1.898	1.914	2.163	2.171	.264	.277	1.8863	1.8997			.127
60	1.875	1.963	1.980	2.238	2.246	.273	.286	1.9515	1.9650	.010	.025	.131
62	1.938	2.029	2.047	2.312	2.321	.282	.295	2.0167	2.0312			.136
64	2.000	2.095	2.113	2.386	2.395	.292	.305	2.0819	2.0819			.140
66	2.062	2.160	2.178	2.459	2.468	.301	.314	2.1472	2.1617			.144
68	2.125	2.226	2.245	2.533	2.543	.310	.323	2.2124	2.2279			.149
70	2.188	2.292	2.311	2.608	2.618	.320	.333	2.2776	2.2931			.153

(D) NOTE: T_T IS THE THEORETICAL INTERSECTION OF THE 144° ± 2° SURFACES. T_T DEFINES THE SIZE OF INSCRIBED HEXAGON BY THE RELATIONSHIP: T_T = HEXAGON SIZE ÷ COS 15°

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PROCUREMENT SPECIFICATION | SUPERSEDES:

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